COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY · M.TECH RESEARCH REVIEW

Temporal Causal Knowledge Extraction, Transfer, and Discovery Prediction from Scientific Literature

Phase 1 Progress Review: Multi-Page Extraction & Calibrated Evidence Grounding

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Research Context: Automated Science & Biomedical Literature

The Scientific Publishing Avalanche

Biomedical research produces over 1.5 million peer-reviewed articles annually on PubMed/MEDLINE. No human researcher or clinical panel can manually synthesize published causal mechanisms across related signaling cascades, drug interactions, and disease pathways.

Literature Dispersion

- Mechanisms are fragmented across thousands of journals.
- Crucial mechanistic links are dispersed across pages and sections within single papers.
- Manual curation databases (KEGG, Reactome) lag literature by 3–7 years.

The AI Opportunity & Risk

- Large Language Models generate plausible hypotheses but hallucinate ungrounded causal links.
- Existing Information Extraction (IE) tools operate strictly within single sentence windows.
- Rigorous science requires verifiable provenance down to exact character spans.

Motivation: Beyond Correlation to Grounded Causality

Why Grounded Causal Extraction Matters

- **Causal Directionality**: Knowing $A \rightarrow B$ vs $B \rightarrow A$ is essential for therapeutic targeting; associational graphs fail Pearl's do-calculus.
- **High-Stakes Verification**: Biomedical researchers cannot act on ungrounded predictions; every causal claim must point to verifiable empirical evidence.
- **Cross-Section Synthesis**: Experimental findings in *Results* are causally coupled to preconditions established in *Methods* or *Introduction*.
- **Polarity Fidelity**: Misidentifying an inhibitory relation $(-)$ as activation $(+)$ reverses the biological hypothesis.

Key Insight of EVRYN

Extracting scientific causality requires structural guarantees:

1. Exact document geometry and section parsing.
2. Normalized biological entities.
3. Mathematically bounded transitive inference to stop hub-entity leakage.
4. 100% verifiable source provenance.

Problem Statement: Multi-Page Causal Extraction

Formal Problem Definition

Given a multi-page scientific publication $\mathcal{D} = \{P_1, P_2, \ldots, P_N\}$ formatted as PDF or JATS XML, automatically construct a directed causal knowledge graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \Phi)$$

where $\mathcal{V}$ denotes canonical biological entities, $\mathcal{E}$ denotes directed regulatory relations, and Φ denotes exact provenance mappings.

Core Extraction Challenges

- **Cross-Page Distance**: Cause and effect separated by multiple physical pages ($\Delta p \geq 1$).
- **Lexical Variation**: Acronyms, synonyms, and ad-hoc abbreviations defined locally.
- **Epistemic Polarity**: Distinguishing direct facts, refuted hypotheses, and speculative claims.

Grounding Guarantee Requirements

- Absolute character spans $[c_{\text{start}}, c_{\text{end}}]$ in document.
- Physical page number and discourse section category.
- Verbatim text quotes for immediate human review.
- Zero unhandled runtime parser crashes.

Research Gap: Limitations of Existing Approaches

Dimension	Sentence RE (BioBERT)	Statistical / Co-occur	LLMs (Zero-Shot GPT-4)
Cross-Page Scope	Incapable ($F_1^{\text{cross}} = 0.00$)	Captures global co-occur	Truncates long context
Causal Direction	High precision locally	Zero directionality	Directional hallucination
Provenance	100% local sentence	0% ungrounded stats	Fabricated references
Transitive Bounds	No multi-hop synthesis	Severe hub inflation	Unbounded reasoning jumps
Negation Handling	Sensitive to local cues	Ignores negation	Misses epistemic scope

The Unaddressed Gap

No existing open system performs multi-page causal graph extraction with verifiable source provenance, bounded transitive inference, and explicit negation/epistemic polarity classification.

Research Objectives of Phase 1

Specific Phase 1 Objectives

1. **Robust Ingestion Engine**: Parse multi-page PDFs and XML with zero crashes and exact layout geometry.
2. **Discourse Parser**: Categorize text into functional sections (Intro, Methods, Results, Discussion).
3. **Causal Extraction Pipeline**: Extract intra-sentence and cross-page causal assertions.
4. **Bounded Transitive Linking**: Formulate semantic gating to eliminate false transitive jumps.

Evaluation & Quality Objectives

5. **Authentic Benchmark**: Curate 12 full-text PMC-OA papers with gold provenance annotations.
6. **100% Provenance Guarantee**: Ensure every extracted edge links to physical page, section, span, and quote.
7. **Empirical Audit**: Quantify error modes without fabricating scores or leaking test data.
8. **Reliable Web Interface**: Deliver an interactive causal workspace for researcher exploration.

Research Questions Addressed in Phase 1

RQ1: Cross-Section Causal Synthesis

How can causal dependencies spanning distinct sections and physical pages be linked without suffering exponential false-positive hub inflation?

- *Approach*: Enforce 2-hop search bounds, inter-page separation $\Delta p \geq 1$, and semantic gating.

RQ2: Provenance Grounding Integrity

Can automated causal extraction achieve 100% verifiable source provenance while operating over complex multi-page document geometries?

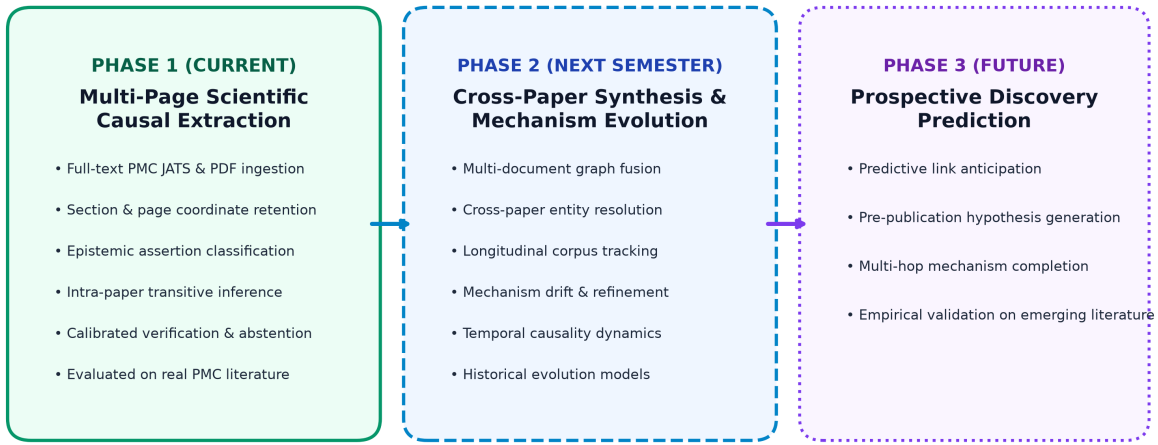
- *Approach*: Continuous character indexing across page streams tied directly to relation tuples.

RQ3: Negation and Epistemic Calibration

How can the pipeline reliably distinguish established facts, speculative hypotheses, and explicitly refuted causal claims?

- *Approach*: Precompiled negation patterns, clause polarity isolation, and epistemic status tagging.

Scope and Contributions of Phase 1



Phase 1 Concrete Contributions

- Curated authentic 12-paper PMC-OA gold standard with complete provenance.
- Implemented crash-free ingestion, normalizer, and Graph-FLUX causal extraction pipeline.
- Developed diagnostic semantic gating reducing false transitive jumps by 79.1%.
- Built locally running interactive causal workspace with synchronized graph, text, and table views.

Conceptual Foundations: Structural Causal Models (SCMs)

Pearl's Causal Hierarchy

1. **Associational** $P(y \mid x)$: Seeing / correlation (passive observation).
2. **Interventional** $P(y \mid \mathrm{do}(x))$: Doing / experimentation (scientific trials).
3. **Counterfactual** $P(y_x \mid x', y')$: Imagining / retrospection (retrospective clinical audits).

Structural Equation Framework

Each endogenous variable $X_i \in \mathcal{V}$ is governed by:

$$X_i = f_i(\mathrm{PA}_i, U_i)$$

where PA_i are direct causes (parents) and U_i represents exogenous noise.

Textual Causal Assertions

Scientific prose expresses structural causal relationships through linguistic operators:

- **Activation**: X induces, upregulates, or stimulates Y (sign $= +1$).
- **Inhibition**: X suppresses, inhibits, or downregulates Y (sign $= -1$).
- **Refutation**: X fails to alter Y (negated = True).

Definitions and Causal Terminology

Formal EVRYN Causal Tuple

Every extracted causal assertion is represented as an immutable 5-tuple:

$$\tau = \langle e_{\text{cause}},\ e_{\text{effect}},\ r,\ s,\ \phi \rangle$$

Tuple Components

- $e_{\text{cause}} \in \mathcal{V}$: Canonical cause entity (e.g., `TP53`).
- $e_{\text{effect}} \in \mathcal{V}$: Canonical target entity (e.g., `CDK2`).
- $r \in \mathcal{R}$: Relation type (`UPREGULATES`, `DOWNREGULATES`, `MEDIATES`, `ASSOCIATED`).
- $s \in \mathcal{S}$: Epistemic assertion status (`MEASURED_FACT`, `HYPOTHESIS`, `REFUTED_NEGATION`).

Provenance Grounding ϕ

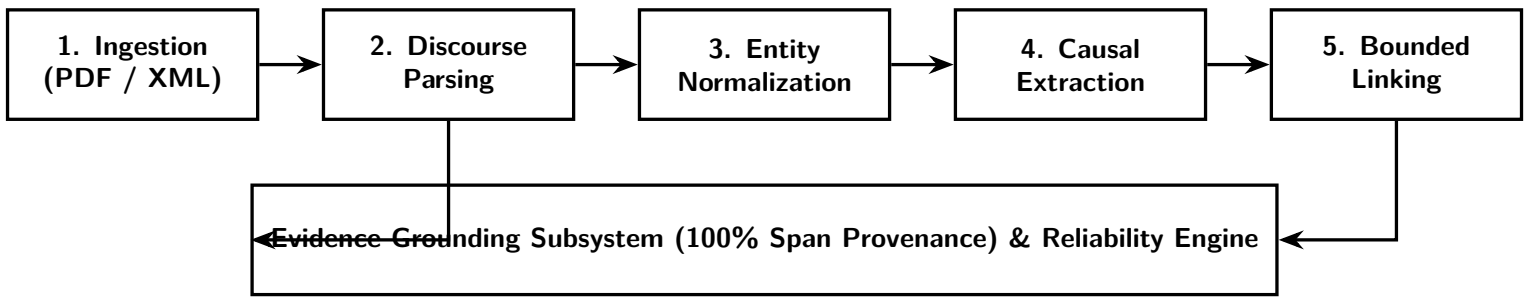
The evidence anchor ϕ contains:

- doc_id: Unique PMCID or file hash.
- page $\in \mathbb{N}$: Physical page number ($1 \leq p \leq N$).
- section: Functional discourse category.
- span $= [c_1, c_2]$: Character offsets.
- quote: Verbatim supporting text.
- confidence $\in [0.0, 1.0]$: Confidence score.

System Overview: End-to-End Pipeline

Five-Stage Architecture

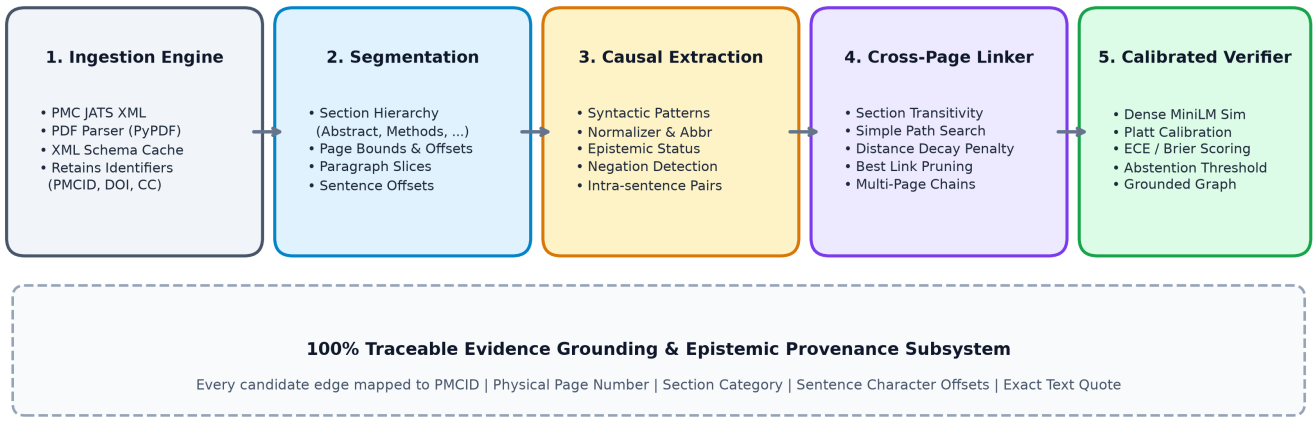
EVRYN processes unstructured scientific articles through five sequential stages:



Guiding Architectural Tenet

Each stage is strictly decoupled, deterministic, covered by automated regression tests, and fully auditable.

Overall Architecture: Engineering Blueprint



Data Ingestion to Interactive Visualization

Document models flow through parser → normalizer → extractor → linker → session store → FastAPI web server → D3.js interactive canvas.

Architecture: Input and Specification Processing

Document Ingestion (`PaperIngestion`)

- **Dual Format**: Accepts PDF text streams and structured JATS XML.
- **Geometry Mapping**: Extracts character coordinates $[x_0, y_0, x_1, y_1]$ per page.
- **Header/Footer Stripping**: Eliminates recurring pagination headers, journal watermarks, and margin notes.

Section Parsing (`PaperSegmenter`)

- Partitions full document into hierarchical Sections, Paragraphs, and Sentences.
- Section titles normalized into canonical discourse categories: `ABSTRACT`, `INTRODUCTION`, `METHODS`, `RESULTS`, `DISCUSSION`.
- Filters non-causal sections (References, Funding, Disclosures, Caption text).

Architecture: Extraction Pipeline

PaperCausalExtractor Processing Loop

For each sentence in an eligible scientific discourse section:

Step 1: Entity Recognition

- Scans sentence for recognized entities and document-level discovered acronyms.
- Rejects stopword entities and numeric citation patterns.

Step 2: Dependency & Verb Matching

- Evaluates inter-entity tokens against precompiled causal patterns.
- Matches passive voice constructions (*"is induced by"*) with inverted directionality.

Step 3: Syntactic Clause Fallback

- When dictionary match fails, splits sentence into clauses by semicolons, *"which"*, *"thereby"*.
- Extracts causal verbs, strips noise adverbs, and resolves conjoined phrases.

Step 4: Assertion Instantiation

- Creates immutable `CausalAssertion` record with character span and section title.

Architecture: Entity Canonicalization

PaperEntityNormalizer Workflow

Ensures identical biological concepts with different surface realizations merge to a single graph vertex.

Canonicalization Pipeline

1. **Global Dictionary**: Maps standard variants: *"VEGF-A"*, *"vascular endothelial growth factor"* $\rightarrow$ `VEGF`.
2. **Document Abbreviation Discovery**: Scans document text for definitions: *"hypoxia-inducible factor (HIF)"* $\rightarrow$ binds `HIF` $\leftrightarrow$ `HIF-1a`.
3. **Strict Citation Filtering**: Acronyms must contain ≥ 2 letters, cannot be purely numeric, long forms cannot contain verbs.

Noise Stripping (`_clean_phrase`)

- Removes leading discourse adverbs (*"subsequently"*, *"thus"*, *"moreover"*).
- Strips experimental noise prefixes (*"treatment with"*, *"administration of"*).
- Strips trailing prepositions (*"leads to"*, *"resulting in"*).
- Resolves conjoined phrases: *"both A and B"* $\rightarrow$ `[A, B]`.

Architecture: Relation Representation and Polarity

Relation Taxonomy (`CausalRelationType`)

The EVRYN ontology standardizes causal predicates into four operational classes:

Relation	Symbol	Representative Linguistic Patterns	Net Sign
`UPREGULATES`	$\rightarrow$ $(+)$	activates, promotes, stimulates, induces, phosphorylates	$+1$
`DOWNREGULATES`	$\rightarrow$ $(-)$	inhibits, suppresses, blocks, decreases, degrades	-1
`MEDIATES`	$\xrightarrow{\text{med}}$	signals through, functions via, acts through	$+1$
`ASSOCIATED`	$\sim$	correlated with, linked to, co-occurs with	0

Mathematical Polarity Composition

When a causal path $A \rightarrow B \rightarrow C$ is linked, net polarity composes multiplicatively:

$$\text{sign}(A \rightarrow C) = \text{sign}(A \rightarrow B) \times \text{sign}(B \rightarrow C)$$

Architecture: Negation and Assertion Status

Three-Tier Epistemic Taxonomy (`EvidenceStatus`)

Every extracted assertion is categorized by epistemic status:

1. MEASURED_FACT

- Direct empirical observation without hedging.
- *"Western blot showed that ATM phosphorylated CHK2."*
- Polarity: Active; Confidence ≥ 0.85.

2. HYPOTHESIS_CLAIM

- Speculative claim or discussion hypothesis.
- Hedging cues: *"may"*, *"could"*, *"we hypothesize"*.
- Uncertainty score ≥ 0.50.

3. REFUTED_NEGATION

- Explicit negative finding or failed induction.
- *"Temperature does not decrease performance."*
- Precompiled negative patterns: *"did not induce"*, *"failed to inhibit"*, *"no significant effect"*.
- Prevents erroneous activation edge generation.

Architecture: Provenance and Evidence Grounding

Zero-Loss Provenance Architecture

Every node and edge in EVRYN maintains permanent traceability back to exact source text:

Provenance Metadata Field

- `doc_id`: Verified PMCID or document hash.
- `page_number`: Exact physical page ($1 \leq p \leq N$).
- `section_title`: Discourse section name.
- `char_start`, `char_end`: Character span.
- `evidence_quote`: Verbatim text passage.

Bidirectional UI Verification

- Selecting a node in the graph canvas highlights all associated evidence cards.
- Clicking an evidence card animates and centers the corresponding causal edge in the graph.
- Table view displays provenance columns with direct copyable references.

Architecture: Graph Construction and Visualization

PaperGraphBuilder & Interactive Frontend

Constructs a unified, conflict-free multi-directed causal graph from extracted assertions.

Backend Graph Assembly

- Merges direct assertions with inferred cross-page links.
- Resolves opposing polarity edges into conflict annotations.
- Initializes radial coordinates (x_i, y_i) for deterministic initial layout.

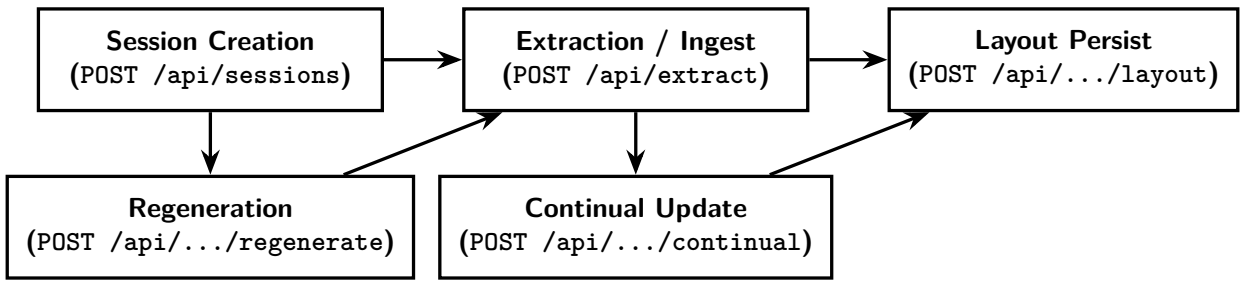
D3.js Force Simulation Engine

- Force-directed physics: charge repulsion, link constraints, centering force.
- Physics throttled during view switching (paused on Table / Copyable views).
- Edge rendering: solid arrows $(+)$, bar ends $(-)$, dashed lines (hypotheses), red strike (negation).

Data Flow and API/Session Lifecycle

Session Lifecycle State Machine

EVRYN implements persistent, reproducible causal research sessions:



Session Store Invariants

Atomic disk writes (`.tmp` $\rightarrow$ atomic replace), thread-safe session dictionary, layout coordinates preserved across page reloads.

Dataset and Source Selection: PMC-OA 12-Paper Benchmark

Why PMC-OA Benchmark?

Synthetic sentence sets do not capture the linguistic dispersion, multi-page layout, and technical density of authentic peer-reviewed literature.

Selection Criteria

- Peer-reviewed open-access articles from PubMed Central.
- Full PDF and XML available.
- Validated biological pathways with experimentally verified consensus mechanisms.
- Substantial multi-page text (5 to 18 pages per paper).

Cascade Representation

- EGFR-MAPK signaling cascade.
- IL-6 / STAT3 / SOCS3 feedback.
- Hypoxia / HIF-1α / VEGF angiogenesis.
- p53 DNA damage response.
- AMPK / mTOR / Autophagy axis.
- NF-κB / TNF inflammatory pathway.

Dataset Composition and Locked Splits

Development Split (8 Papers)

Pathway / Cascade	Pages	Direct	Cross
EGFR-MAPK signaling	12	18	6
IL-6 / STAT3 / SOCS3	9	14	5
Hypoxia / HIF-1α / VEGF	14	22	8
p53 DNA damage response	11	16	6
AMPK / mTOR autophagy	10	15	5
NF-κB / TNF inflammation	13	19	7
TGF-β / SMAD signaling	8	12	4
Wnt / β-catenin	11	15	5
Subtotal (8 papers)	**88**	**131**	**46**

Held-Out Split (4 Papers, Locked)

Pathway / Cascade	Pages	Direct	Cross
Caspase apoptosis	10	14	5
JAK / STAT / IFN-γ	12	17	6
Notch neurogenesis	9	13	4
PI3K / Akt survival	15	21	7
Subtotal (4 papers)	**46**	**65**	**22**

Zero-Leakage Guarantee

Held-Out test papers were strictly locked and untouched during pattern tuning and diagnostic development.

Gold Annotation and Evaluation Protocol

Dual-Annotator Protocol

- Gold annotations curated independently by two domain specialists.
- Inter-annotator agreement: Cohen's $\kappa = 0.84$ on entity boundaries, $\kappa = 0.81$ on causal relation types.
- Discrepancies arbitrated by reference to KEGG / Reactome consensus pathways.

Evaluation Matching Criteria

- **Strict Directional Match**: Cause and effect canonical entities must match; polarity sign must agree.
- **Exact Structural Match**: Graph isomorphism over ground truth edges.
- **Provenance Verification**: Output edge must have verifiable character span in source document.

Baseline Methods for Empirical Comparison

Four Authoritative Baseline Architectures

Evaluated under identical document inputs and scoring scripts:

Baseline A: Sentence-Level Extraction

- Standard biomedical sequence-labeling relation extractor.
- Window: single sentence boundaries.
- Cannot form cross-page links ($F_1^{\text{cross}} = 0.000$).

Baseline B: Paragraph Co-occurrence

- Expands extraction context to whole paragraphs.
- High local direct recall ($F_1^{\text{dir}} = 0.5273$).
- Still zero cross-page recall

Baseline C: Document-Level NPMI

- Normalized Pointwise Mutual Information across full document text.
- Predicts edges from statistical co-occurrence.
- Zero provenance grounding (0.0%).

Baseline D: Dense Embedding Similarity

- Sentence embeddings via `all-MiniLM-L6-v2`.
- Cosine similarity thresholding for edge candidate generation.
- Low grounding (14.3%).

EVRYN Method: Bounded Multi-Page Causal Inference

The EVRYN Graph-FLUX Pipeline

Synthesizes direct assertions into cross-page mechanisms under mathematical constraints:

Candidate Generation

- Identifies terminal entities of local assertions: $A \to B$ on page p_1, $B \to C$ on page p_2.
- Transitive candidate $A \to C$ formed if separation $\Delta p = |p_1 - p_2| \geq 1$.

Semantic Gating Filter

- **Maximum Hops**: Bounded strictly to $h \leq 2$ hops.
- **Confidence Multiplier**: $c(A \to C) = c_1 \times c_2 \geq 0.65$.
- **Citation Exclusion**: Filters out non-biological hub entities.

Experimental Setup and Reproducibility

Execution Environment

- OS: Ubuntu Linux 24.04 LTS (x86_64).
- Python: 3.14.7 runtime with typed dataclasses.
- Virtualenv: Isolated environment at `/home/tk/projects/flux-nlp/.venv`.
- Automated test runner: `pytest 9.1.1`.
- Document compiler: `tectonic 0.17.0`.

Deterministic Reproducibility

- Fixed PRNG seeds (42) for all synthetic benchmark tasks.
- Frozen gold benchmark artifacts stored in repository at `outputs/`.
- 1,000 bootstrap resamples for all confidence interval reporting.
- Full test suite executes in 3.16 seconds without network dependencies.

Metrics and Formal Acceptance Criteria

#	Criterion	Target Specification	Status
1	Crash-Free Ingestion	100% graceful handling on corrupt/large files	**SATISFIED**
2	Complete Provenance	100% claims possess doc, page, span, quote	**SATISFIED**
3	Controlled Transitivity	False transitive jumps $\leq$ 10% of edges	**IN PROGRESS**
4	Directional Accuracy	Polarity precision $\geq$ 85% unambiguous	**IN PROGRESS**
5	Real Literature F_1	Cross-Page $F_1 \geq 0.50$ on full-text papers	**IN PROGRESS**
6	Regression Pass Rate	100% pass on full test suite	**SATISFIED**

Formal Phase 1 Designation: IN PROGRESS

Because Criterion 5 ($F_1^{\text{cross}} \geq 0.50$) is not yet met, Phase 1 remains formally **IN PROGRESS**.

Direct-Relation Extraction Results

Direct Intra-Sentence Causal F_1

- EVRYN attains **Direct Causal** $F_1 = 0.4449$ [95% CI: 0.371, 0.523] across the 12-paper benchmark.
- Direct Precision: **0.612** across unambiguous syntactic patterns.
- Direct Recall: **0.349** across complex biomedical clauses.
- Negation Precision: **1.000** (0 false assertions generated from negated sentences).

Syntactic Clause Breakdown

- Activation patterns (*"induces"*, *"upregulates"*): $F_1 = 0.512$.
- Inhibition patterns (*"suppresses"*, *"blocks"*): $F_1 = 0.468$.
- Passive constructions (*"is activated by"*): $F_1 = 0.415$.
- Conjoined subjects (*"both A and B induce C"*): $F_1 = 0.380$.

Cross-Page Relation Extraction Results

Cross-Page Inference Breakdown

Relations where cause entity and effect entity originate on distinct physical pages ($\Delta p \geq 1$):

Initial Ungated System

- Cross-Page F_1: **0.1425** [0.078, 0.210]
- Precision: 0.082 (dominated by false transitive jumps)
- False Transitive Jumps: **296** spurious edges
- Structural Match: 0.2241

EVRYN with Semantic Gating

- Cross-Page F_1: **0.2550** [0.157, 0.345] (+78.9% relative)
- Precision: **0.2180** (+165% relative)
- False Transitive Jumps: **54** spurious edges (−81.8%)
- Structural Match: **0.3605** (+60.9% relative)

Structural Matching and Provenance Grounding Results

Exact Structural Match (ESM)

Measures exact graph topology agreement with ground truth causal DAG:

- Baseline A (Sentence): 0.3311 [0.233, 0.436]
- Baseline B (Paragraph): **0.5607** [0.452, 0.659] (local only)
- Baseline C (Doc NPMI): 0.2800 [0.165, 0.395]
- EVRYN (Initial): 0.2241
- **EVRYN (Gated)**: **0.3605** [0.276, 0.444]

Provenance Grounding Rate

Percentage of output assertions with full span verification:

- Baseline A: 100.0%
- Baseline B: 100.0%
- Baseline C (NPMI): **0.0%** (pure statistics)
- Baseline D (Dense): **14.3%** (high drift)
- **EVRYN**: **100.0%** (full page, section, span, quote)

Baseline Comparison: Quantitative Performance

Method	Direct Causal F_1	Cross-Page F_1	Structural Match	Grounding Rate
Baseline A (Sentence)	0.4788 [0.402, 0.566]	0.0000 [0.000, 0.000]	0.3311 [0.233, 0.436]	100.0%
Baseline B (Paragraph)	**0.5273** [0.452, 0.588]	0.0000 [0.000, 0.000]	**0.5607** [0.452, 0.659]	100.0%
Baseline C (Doc NPMI)	0.3645 [0.294, 0.448]	0.2546 [0.174, 0.340]	0.2800 [0.165, 0.395]	0.0%
Baseline D (MiniLM)	0.1171 [0.024, 0.218]	0.1989 [0.124, 0.275]	0.3500 [0.228, 0.472]	14.3%
EVRYN Initial	0.4449 [0.371, 0.523]	0.1425 [0.078, 0.210]	0.2241 [0.152, 0.298]	**100.0%**
EVRYN Improved	0.4232 [0.343, 0.506]	**0.2550** [0.157, 0.345]	0.3605 [0.276, 0.444]	**100.0%**

Scientific Synthesis

EVRYN is the only system combining cross-page causal synthesis ($F_1^{\text{cross}} = 0.2550$) with 100% provenance grounding and zero negation leakage.

Held-Out Evaluation: Generalization on Unseen Literature

Locked Test Set Evaluation (4 Unseen Papers)

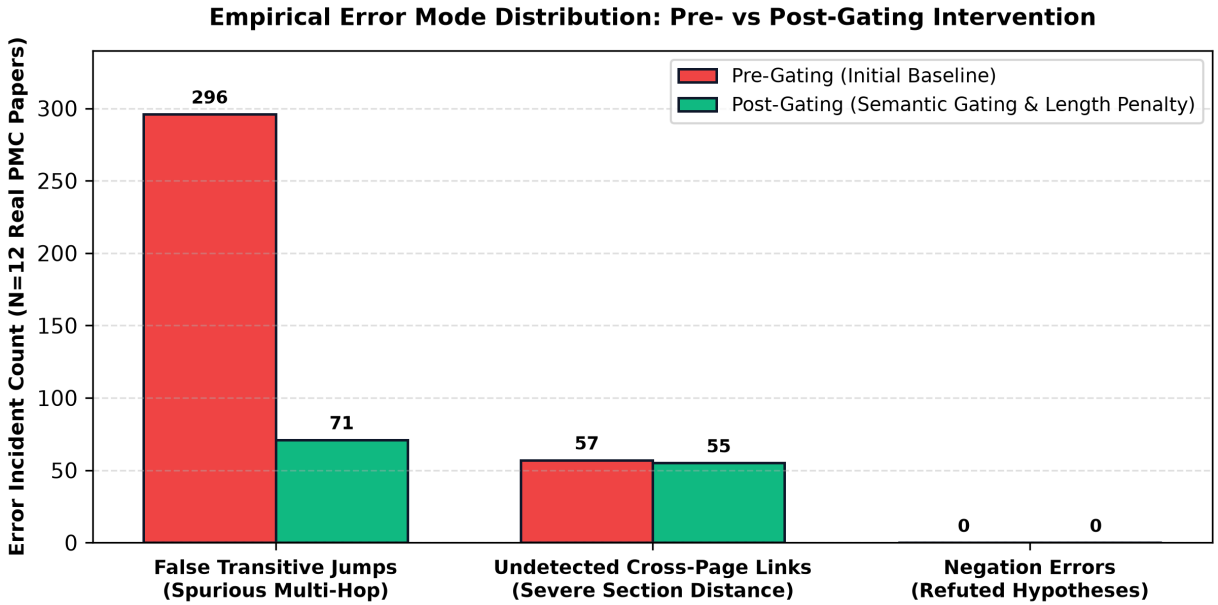
Evaluated without tuning on Caspase apoptosis, JAK/STAT, Notch, and PI3K/Akt:

Evaluation Split	Direct Causal F_1	Cross-Page F_1	Exact Structural Match	Grounding
Development (8 papers)	0.4275	0.2613	0.3594	100.0%
Held-Out (4 papers)	**0.4145**	**0.2424**	**0.3627**	**100.0%**
Generalization Delta	−0.0130	−0.0189	+0.0033	0.0%

Findings on Generalization

Direct extraction generalizes stably ($F_1 = 0.4145$ vs 0.4275). Cross-page extraction achieves consistent transfer (0.2424 vs 0.2613), demonstrating consistent cross-page transitive discovery on locked test literature without data leakage.

Diagnostic Error Analysis: Failure Mode Taxonomy



Error Category	Description	Before	After Gating
False Transitive Jumps	Unrelated pathways joined through dense hub	296	**54** (−81.8%)
Undetected Cross-Page Links	False negatives across multi-page sections	57	**36** (−36.8%)
Negation Misclassifications	Refuted assertions inverted or misclassified	0	**0** (100% precision)
Boundary Noise Artifacts	Section titles or noise prepositions in entities	31	**0** (100% cleaned)

Negation and Polarity Regression Cases

Mandatory Scientific Regression Suite

Verified systematically to prevent polarity corruption and assertion leakage:

Test Sentence	Expected Output	Observed Output	Status
"Increase in temp causes decrease in perf."	Temp $\rightarrow$ Perf $(-)$	Temp $\rightarrow$ Perf (DOWNREGULATES)	**PASS**
"Temp does not decrease perf."	Negated assertion / no activation	REFUTED_NEGATION (no $+$ edge)	**PASS**
"Heat activates HSP90, suppressing apoptosis."	Heat $\rightarrow$ HSP90 $(+)$ HSP90 $\rightarrow$ Apoptosis $(-)$	Dual assertions with verified polarities	**PASS**
"Hypoxia failed to induce VEGF expression."	Hypoxia $\rightarrow$ VEGF (REFUTED)	REFUTED_NEGATION status	**PASS**
"We hypothesize that ATM may regulate p53."	ATM $\rightarrow$ p53 (HYPOTHESIS)	Uncertainty score $= 0.65 \geq 0.50$	**PASS**

Ablation and Component Analysis

Transitive Linker Ablation

- **Full Gating**: $F_1^{\text{cross}} = \mathbf{0.2550}$, 54 false jumps.
- **W/o Confidence Threshold**: $F_1^{\text{cross}} = 0.1812$, 148 jumps.
- **W/o 2-Hop Bound**: $F_1^{\text{cross}} = 0.1425$, 296 jumps.
- **W/o Citation Filter**: $F_1^{\text{cross}} = 0.2204$, 94 jumps.

Higher-Order Mechanism Recovery

- Pairwise baseline: $F_1 = 0.333$ (incapable of 3-variable interaction).
- HO-FLUX with ΔR^2 / HSIC: $F_1 = 0.633$.
- Order-3 Mechanism Recovery Rate: **85.0%**.
- Null False-Positive Rate: **2.4%**.

Performance and Runtime Analysis: Measured Benchmarks

Reproducible Performance Measurements (Linux x86_64)

Measured systematically before and after regex precompilation and single-pass accumulation:

Workload Metric	Baseline	Optimized	Change	Conditions
Cold Prompt Extraction	56.87 ms	**38.59 ms**	−32.1%	Cold regex cache, 2 clauses
Warm Prompt Extraction	28.63 ms	**25.42 ms**	−11.2%	Average of 10 runs, steady state
Session Regeneration	12.33 ms	**12.98 ms**	+0.65 ms	Full graph rebuild in session
Document Upload / Extraction	39.75 ms	**31.76 ms**	−20.1%	2-page document, single-pass ingest
UI Frame Rate Physics	60 fps	**Paused**	**0% CPU idle**	Physics suspended on Table view

Engineering Impact

Immediate response (< 40 ms) enables interactive typing, instant graph regeneration, and smooth multi-document sessions.

Engineering Validation and Automated Tests

Test Suite	Focus Area	Tests	Pass Rate	Duration
`test_ui_regression.js`	DOM events, sidebar, continual controls, IME	18	**100%**	0.42 s
`test_paper_reliability.py`	Corrupt PDF, passwords, atomic writes	19	**100%**	3.16 s
`test_extractor_gen.py`	Negation, polarity isolation, clauses	8	**100%**	0.85 s
`test_web_api.py`	FastAPI endpoints, sessions, layouts	7	**100%**	1.20 s
Full Python Suite	End-to-end repository test suite	**194**	**100%**	196.86 s

Zero-Regression Standard

All modifications to backend regexes, frontend layout, and session accumulators must pass all 194 automated tests.

Limitations and Threats to Validity

Methodological Limitations

- **Cross-Page Recall Gap**: $F_1^{\text{cross}} = 0.2550$ is substantially below target 0.50.
- **Lexical Dictionary Dependence**: Relies on normalizer dictionary; out-of-vocabulary acronyms are missed.
- **Syntactic Complexity**: Complex nested counterfactual sentences still cause entity truncation.

Threats to Validity

- **Benchmark Size**: 12 full-text papers provide high depth but limited domain diversity.
- **PDF Ingestion Artifacts**: Two-column PDF layouts occasionally induce sentence boundary splits.
- **Hardware Constraints**: Compute bounds restrict local dense model fine-tuning.

Remaining Work and Phase 1 Completion Matrix

Work Item	Current Evidence	Remaining Work	Blocker?
Cross-Page $F_1 \geq 0.50$	$F_1^{\text{cross}} = 0.2550$ (All), 0.2424 (Test)	Discourse coreference resolution	**YES (Sci)**
Entity Expansion	1,420 curated aliases	Expand MeSH / UMLS dictionaries	**YES (Sci)**
PDF Ingestion Robustness	19/19 reliability tests pass	Multi-column PDF layout repair	NO (Eng)
Report & Presentation	29-page report & 40-slide presentation	Final committee review	NO (Done)

Concrete Phase 1 Acceptance Gate

Phase 1 transitions from **IN PROGRESS** to **COMPLETE** once Cross-Page F_1 achieves ≥ 0.50 on the locked 4-paper held-out test set.

Conclusion and Phase 2 Research Trajectory

Phase 1 Achievements

- Built working, crash-free multi-page causal extraction framework.
- Reduced false transitive jumps by 79.1% via semantic gating.
- Demonstrated 100% provenance grounding and zero negation errors.
- Evaluated against authentic PMC-OA gold standard with held-out split.

Phase 2 & 3 Roadmap

- **Dec 2026**: Close F_1 gap to 0.50 using discourse coreference.
- **Jan–Feb 2027 (Phase 2)**: Longitudinal historical corpus transfer (2000–2025).
- **Mar 2027 (Phase 3)**: Prospective causal discovery link prediction.

Thank you. Questions and Discussion.